

LAB07_IsaDias

Isabella Ferreira Dias 266903

Atividade

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(DBI)
library(RSQLite)

path <- getwd()
fname <- file.path(path, "database.sqlite3")

conn <- dbConnect(SQLite(), fname)
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
sql <- "
SELECT DISTINCT instructors.name AS PROFESSOR FROM instructors
INNER JOIN teachings ON instructors.id = teachings.instructor_id
INNER JOIN sections ON teachings.section_uuid = sections.uuid
INNER JOIN course_offerings ON sections.course_offering_uuid = course_offerings.uuid
INNER JOIN subject_memberships ON course_offerings.uuid = subject_memberships.course_offering
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'
ORDER BY PROFESSOR;
"

professores_stat <- dbGetQuery(conn, sql)
```

professores_stat

	PROFESSOR
1	ABIGAIL BENZINE
2	ADIN-CRISTIAN ANDREI
3	AKICHIKA OZEKI
4	ALAN HUANG
5	ALBERTO DEL PIA
6	ALBRECHT KLEMM
7	ALEXANDER COVINGTON
8	ALEXANDER FISH
9	ALEXANDER KISELEV
10	ALYSSA DIGILIO
11	ANATOLE BECK
12	ANDREJ ZLATOS
13	ANDREW LESLIE
14	ANNE BRUCKNER
15	ANQI SHI
16	ANRU ZHANG
17	ARNOLD MILLER
18	AUGUST JENSEN
19	BEHZAD AALIPUR
20	BEHZAD AALIPUR HAFSHEJANI
21	BEN ADAM HAALAND
22	BENEDEK VALKO
23	BENJAMIN RECHT
24	BI CHENG WU
25	BIN DAI
26	BIN ZHANG
27	BIN ZHU
28	BINGYING XIE
29	BO HUANG
30	BO YANG
31	BOWEN HU
32	BRET HANLON
33	BRET LARGET
34	BRIAN YANDELL
35	BROOK LUERS
36	CECILE ANE
37	CHAN PARK
38	CHANHAN HSU

39	CHAOQUN MEI
40	CHAOYANG YU
41	CHELSEY GREEN
42	CHEN CHENG
43	CHEN JING
44	CHENGNING ZHANG
45	CHENLIANG XU
46	CHENSHENG KUANG
47	CHENXI LI
48	CHIA-CHIEH LIN
49	CHIEN-WEI CHEN
50	CHRISTINA M. KENDZIORSKI
51	CHRISTINE SORKNESS
52	CHRISTOPHER WAGNER
53	CLAIRE BOBST
54	CLAUDIA SOLIS LEMUS
55	COLE COOK
56	COLIN LONGHURST
57	CRYSTAL CHEN
58	CUICUI QI
59	CUIZE HAN
60	DANIELE CAPPELLETTI
61	DAVID ANDERSON
62	DAVID DEMETS
63	DAVID GRIFFEATH
64	DEBRAJ DAS
65	DEREK BEAN
66	DEREK NORTON
67	DEYUAN JIANG
68	DIETRICH UHLENBROCK
69	DONALD PORTER
70	DONG XIA
71	DONGGYU KIM
72	DONGHYUN LEE
73	DOUGLAS M. BATES
74	DUY NGUYEN
75	DUZHE WANG
76	EDWARD ERKER
77	ELOISA D CHAVAS
78	ERICA LEE DEADMAN
79	ERIK NORDHEIM
80	FAN GAO
81	FAN YANG

82	FANG FANG
83	FANGFANG WANG
84	FLORIAN BERTRAND
85	FREDERICK BOEHM
86	GARVESH RASKUTTI
87	GARY HOWARD SCHROEDER
88	GENG LI
89	GINA BENNINGER
90	GINA OH
91	GONZALO CONTADOR
92	GRACE WAHBA
93	GREGORIO MORENO-FLORES
94	GREGORY SHINAULT
95	GUANHUA CHEN
96	GUANNAN SUN
97	GUILHERME ROSA
98	GUILHERME VIEIRA NUNES LUDWIG
99	GUN WOONG PARK
100	HAN CHEN
101	HAO CHEN
102	HAO TENG
103	HAO ZHENG
104	HAO ZHOU
105	HAODA FU
106	HAOYANG FAN
107	HEATHER MARIE BRAZEAU
108	HUAIBAO FENG
109	HUI WANG
110	HUIKUN ZHANG
111	HUILIN HU
112	HYEBIN SONG
113	HYUNSEUNG KANG
114	IAN BRANSTAD RILEY
115	ISMOR FISCHER
116	JAMES ANDERSON
117	JAMES D KUELBS
118	JARED BROWN
119	JASON P FINE
120	JASON RICHARD SWANSON
121	JEE YEON KIM
122	JEEA CHOI
123	JENNIFER BIRSTLER
124	JENNIFER NGUYEN

125	JESSE THOMAS HOLZER
126	JIAJIE CHEN
127	JIALE XU
128	JIAN WU
129	JIANCHANG HU
130	JIE SONG
131	JIE WEI
132	JIE ZHANG
133	JILI WANG
134	JINGCI MENG
135	JINGJIANG PENG
136	JINGLAN LI
137	JIWEI ZHAO
138	JOHN DAVIS
139	JOHN GILLET
140	JOHN KANE
141	JOHN WILTSHIRE-GORDON
142	JONATHON PETERSON
143	JOSEP GINEBRA
144	JOSEPH DEUTSCH
145	JU HEE CHO
146	JUN LI
147	JUN SHAO
148	JUN YIN
149	JUN ZHANG
150	JUN ZHU
151	JUNGWON MUN
152	JUNHEE HAN
153	JUNHO LEE
154	KAILEI CHEN
155	KAM-WAH TSUI
156	KARL BROMAN
157	KARL ROHE
158	KATHERINE GOODE
159	KAZUHIKO SHINKI
160	KEEGAN KORTHAUER
161	KEN ONO
162	KEVIN HASEGAWA ENG
163	KEVIN PACKARD
164	KJELL DOKSUM
165	KRISHNAKUMAR BALASUBRAMANIAN
166	KRISTEN CYFFKA
167	KUNLING HUANG

168	KYLE HEBERT
169	KYLE HERBET
170	KYUNGMA NN KIM
171	LAM HO
172	LAN LUO
173	LANCINE KONATE
174	LEI XU
175	LEV BORISOV
176	LIAM JOHNSTON
177	LIE XIONG
178	LILI LAN
179	LILI ZHENG
180	LILUN DU
181	LIN QI
182	LONG PHAN
183	LU MAO
184	LU YANG
185	LUWAN ZHANG
186	LUXI CAO
187	MANJUSHA KANCHARLA
188	MARI PALTA
189	MARIA KAMENETSKY
190	MARIA ROJO
191	MARIAN R FISHER
192	MARY LINDSTROM
193	MATTHEW BALLARD
194	MENG SONG
195	MICHAEL FERRIS
196	MICHAEL GEORGE I LTIS
197	MICHAEL HOGAN
198	MICHAEL KUTZLER
199	MICHAEL LIOU
200	MICHAEL NEWTON
201	MICHAEL RENE KOSOROK
202	MICHELLE HARRIS
203	MIHAELA IFRIM
204	MIN JUNG LEE
205	MIN NIU
206	MING XIE
207	MING YUAN
208	MINJING TAO
209	MITCHELL PAUKNER
210	MOO K CHUNG

211	MUHONG GAO
212	MURRAY CLAYTON
213	NATALIA DE LEON GATTI
214	NELLIE LAUGHLIN
215	NICHOLAS HENDERSON
216	NICHOLAS STEPHEN KEULER
217	NING FAN
218	NORBERT BINKIEWICZ
219	NORMAN DRAPER
220	PAUL M TERWILLIGER
221	PAUL RATHOUZ
222	PAUL SAVARIAPPAN
223	PEIGEN ZHOU
224	PERLA REYES
225	PHILIP WOOD
226	QI JIANG
227	QI TANG
228	QIAN ZHIGUANG
229	QING LI
230	QIONG ZHANG
231	QIURONG CUI
232	QUEFENG LI
233	QUOC TRAN
234	REBECCA KOSCIK
235	REBECCA POST
236	RICHARD A BRUALDI
237	RICHARD A. JOHNSON
238	RICHARD J. CHAPPELL
239	ROBERT HARRON
240	ROBERT R MEYER
241	ROBERT WARDROP
242	ROBERT WAYNE GREEN
243	RONALD GANGNON
244	RONGJUN ZHU
245	RUI CHEN
246	RUI TANG
247	RUIFANG SONG
248	RUIFENG XU
249	RUNGANG HAN
250	RUOSI GUO
251	RYAN ZEA
252	SAMUEL STECHMANN
253	SANGBUM CHOI

254	SCOTT HOTTOVY
255	SCOTT JOSEPH HETZEL
256	SEAN KENT
257	SEBASTIEN ROCH
258	SEHO PARK
259	SEULKEE YUN
260	SEUNGBONG HAN
261	SHAN LU
262	SHANE HUBLER
263	SHANG WU
264	SHENG WANG
265	SHENG ZHANG
266	SHENGJI JIA
267	SHI JIN
268	SHIXUE LIU
269	SHIZHEN WANG
270	SHUAI CHEN
271	SHUANG HUANG
272	SHULEI WANG
273	SHUYUN YE
274	SIJIAN WANG
275	SIJING LI
276	SOHEIL SADEGHI
277	SOKOL VAKO
278	SONA Z SWANSON
279	SONG WANG
280	SRIKANTHMADHAVAN ARAVAMUTHAN
281	STEFFEN LEMPP
282	STEPHEN AARON STANHOPE
283	STEPHEN BERG
284	STEPHEN WAINGER
285	STEPHEN WRIGHT
286	STEVEN SAM
287	SUKHENDU MEHROTRA
288	SUNDUZ KELES
289	TAERI UHM
290	TAO YU
291	THEVAASIINEN CHANDERENG
292	THOMAS COOK
293	THOMAS G. KURTZ
294	THU LE
295	TIEN VO
296	TIMO SEPPALAINEN

297	TIMOTHY IDOWU
298	TING YE
299	TING-LI LIN
300	TONG LI
301	TONGHAI YANG
302	TRAM TA
303	TUN LEE NG
304	TZU HSIANG HUNG
305	VARSHA KULKARNI
306	VICTOR LUO
307	VICTORIA MANSFIELD
308	WAI TONG FAN
309	WEI ZHENG
310	WEI-YIN LOH
311	WENWEN ZHANG
312	WENZHI CAO
313	WESLEY CHANG
314	XIAO GUO
315	XIAO NIE
316	XIAODAN WEI
317	XIAOMAO LI
318	XIAOPING FENG
319	XIAOWU DAI
320	XIN LI
321	XIN ZHANG
322	XINJIE HE
323	XINWEI DENG
324	XINXIN YU
325	XINYU SONG
326	XIPEI YANG
327	XIRAN WANG
328	XIUFENG SHAO
329	XIUYU MA
330	XIWEN MA
331	XU HE
332	XU XU
333	XUEYAO CHEN
334	XUN ZHAO
335	YAJUAN SI
336	YALI WANG
337	YAN CHEN
338	YANBING ZHENG
339	YANG ZHAO

340	YANNAN QIU
341	YAOGUO XIE
342	YAOYAO XU
343	YAZHEN WANG
344	YI CHAI
345	YI LI
346	YI LIU
347	YIFAN MEI
348	YILIN ZHANG
349	YING ZHANG
350	YONGFENG WU
351	YONGJOON KIM
352	YONGSU LEE
353	YOUNG LEE
354	YOUNG MIN PARK
355	YOUNGDEOK HWANG
356	YOURAN QI
357	YU MENGANG
358	YU QIU LIU
359	YUAN JIANG
360	YUAN LI
361	YUAN WANG
362	YUANZHI LI
363	YUCHANG WU
364	YUCHEN ZHOU
365	YUJIN CHUNG
366	YUNONG LIN
367	YUQING XU
368	ZHANG CHUNMING
369	ZHENGJUN ZHANG
370	ZHENGXIAO WU
371	ZHIGENG GENG
372	ZHIGUO XIAO
373	ZHONGJIE YU
374	ZHUANG WU
375	ZIFENG ZHAO
376	ZIJIAN NI
377	ZUOFENG SHANG

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

Quem é o professor mais difícil?

```
sql <- "SELECT instructors.name AS PROFESSOR,
ROUND(AVG(
(4.0*gd.a_count +
3.5*gd.ab_count +
3.0*gd.b_count +
2.5*gd.bc_count +
2.0*gd.c_count +
1.0*gd.d_count +
0.0*gd.f_count)*1.0 /

(gd.a_count + gd.ab_count + gd.b_count +
gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
),3) AS GPA_MEDIO

FROM grade_distributions gd

INNER JOIN course_offerings co ON gd.course_offering_uuid = co.uuid
INNER JOIN sections s ON co.uuid = s.course_offering_uuid

INNER JOIN teachings t ON s.uuid = t.section_uuid

INNER JOIN instructors ON t.instructor_id = instructors.id

INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid

INNER JOIN subjects ON sm.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY instructors.id, instructors.name

HAVING GPA_MEDIO IS NOT NULL
```

```
ORDER BY GPA_MEDIO ASC
```

```
LIMIT 1
```

```
"
```

```
dbGetQuery(conn, sql)
```

```
      PROFESSOR GPA_MEDIO  
1 JAMES D KUELBS      2.599
```

Quem é o professor mais fácil?

```
sql <- "SELECT instructors.name AS PROFESSOR,
```

```
ROUND(AVG(
```

```
(4.0*gd.a_count +
```

```
3.5*gd.ab_count +
```

```
3.0*gd.b_count +
```

```
2.5*gd.bc_count +
```

```
2.0*gd.c_count +
```

```
1.0*gd.d_count +
```

```
0.0*gd.f_count)*1.0 /
```

```
(gd.a_count + gd.ab_count + gd.b_count +
```

```
gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
```

```
),3) AS GPA_MEDIO
```

```
FROM grade_distributions gd
```

```
INNER JOIN course_offerings co ON gd.course_offering_uuid = co.uuid
```

```
INNER JOIN sections s ON co.uuid = s.course_offering_uuid
```

```
INNER JOIN teachings t ON s.uuid = t.section_uuid
```

```
INNER JOIN instructors ON t.instructor_id = instructors.id
```

```
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
```

```
INNER JOIN subjects ON sm.subject_code = subjects.code
```

```

WHERE subjects.abbreviation = 'STAT'

GROUP BY instructors.id, instructors.name

HAVING GPA_MEDIO IS NOT NULL

ORDER BY GPA_MEDIO DESC

LIMIT 1
"

dbGetQuery(conn, sql)

```

	PROFESSOR	GPA_MEDIO
1	SRIKANTHMADHAVAN ARAVAMUTHAN	4

Qual é a disciplina mais difícil?

```

sql <- "SELECT courses.name AS DISCIPLINA,

ROUND(AVG(
(4.0*gd.a_count +
3.5*gd.ab_count +
3.0*gd.b_count +
2.5*gd.bc_count +
2.0*gd.c_count +
1.0*gd.d_count +
0.0*gd.f_count)*1.0 /

(gd.a_count + gd.ab_count + gd.b_count +
gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
),3) AS GPA_MEDIO

FROM grade_distributions gd

INNER JOIN course_offerings co ON gd.course_offering_uuid = co.uuid

INNER JOIN courses ON co.course_uuid = courses.uuid

```

```

INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid

INNER JOIN subjects ON sm.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY courses.uuid, courses.name

HAVING GPA_MEDIO IS NOT NULL

ORDER BY GPA_MEDIO ASC

LIMIT 1
"

dbGetQuery(conn, sql)

```

	DISCIPLINA	GPA_MEDIO
1	Introduction to the Theory of Probability	2.892

Qual é a disciplina mais fácil?

```

sql <- "SELECT courses.name AS DISCIPLINA,

ROUND(AVG(
(4.0*gd.a_count +
3.5*gd.ab_count +
3.0*gd.b_count +
2.5*gd.bc_count +
2.0*gd.c_count +
1.0*gd.d_count +
0.0*gd.f_count)*1.0 /

(gd.a_count + gd.ab_count + gd.b_count +
gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
),3) AS GPA_MEDIO

FROM grade_distributions gd

INNER JOIN course_offerings co ON gd.course_offering_uuid = co.uuid

```

```

INNER JOIN courses ON co.course_uuid = courses.uuid

INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid

INNER JOIN subjects ON sm.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY courses.uuid, courses.name

HAVING GPA_MEDIO IS NOT NULL

ORDER BY GPA_MEDIO DESC

LIMIT 1
"

dbGetQuery(conn, sql)

```

	DISCIPLINA	GPA_MEDIO
1	Nonparametric Statistics and Machine Learning Methods	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

Horario

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-21 09:20:58 -03"
```